

Payment & Financial Services - Document 20 - Page 1/5

Document ID: PAY-020 | Classification: Internal Use

Error code 5003: Payment gateway connection timeout. Check network connectivity and gateway health.

Payment gateway requires secure token handling. Never store raw card numbers. Use payment tokens.

Payment processing performance monitoring: latency metrics, throughput, success rates, and SLA thresholds.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

Payment & Financial Services - Document 20 - Page 2/5

Document ID: PAY-020 | Classification: Internal Use

Error code 4001: Invalid payment method. Verify card details and expiry. Support 3DS if required.

EU region mandates PSD2 and Strong Customer Authentication (SCA) for card payments.

Payment database management: transaction logs, audit trails, backup procedures, and data retention policies.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

Payment & Financial Services - Document 20 - Page 3/5

Document ID: PAY-020 | Classification: Internal Use

Error code 4002: Insufficient funds. Customer must use alternative payment method or add funds.

Error 5003 indicates gateway timeout. Typical causes: network issues, gateway overload, DNS failure.

Payment gateway configuration for EU region: PSD2 compliance, SCA requirements, and GDPR data handling.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
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Payment & Financial Services - Document 20 - Page 4/5

Document ID: PAY-020 | Classification: Internal Use

EU payment failures: Verify PSD2 compliance, SCA completion, and issuer 3DS configuration.

Connection pooling for payment DB: set max_connections=50, connection_timeout=30s.

Payment user authentication: 3DS2 flows, biometric verification, and risk-based authentication.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging.
For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
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Payment & Financial Services - Document 20 - Page 5/5

Document ID: PAY-020 | Classification: Internal Use

Intermittent payment failures: Review gateway logs, check rate limits, verify certificate expiry.

For INR to USD discrepancy: verify FX rate source, conversion timestamp, and rounding rules.

Payment transaction logging: structured logs, trace IDs, and observability for incident investigation.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.